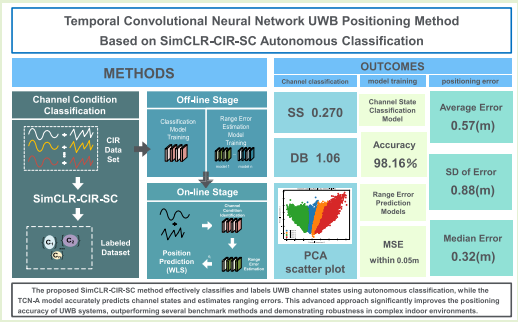
Temporal Convolutional Neural Network UWB Positioning Method Based on SimCLR-CIR-SC Autonomous Classification

Shixun Wu, Xiao Wang, Miao Zhang, *Member, IEEE*, Zheng Chu, *Member, IEEE*, Zhangli Lan, Kai Xu, and Shuang Jin

***Abstract*—Ultrawideband (UWB) technology has garnered considerable research interest for indoor positioning applications owing to its temporal resolution and strong signal penetration capabilities. However, the traditional classification of line-of-sight (LoS) or non-line-of-sight (NLoS) channel condition fails to characterize the complex indoor environments, and the existing models are trained on the entire dataset without considering the heterogeneity of the data, resulting in suboptimal positioning accuracy and reliability. To overcome these limitations, we propose an innovative autonomous classification method called SimCLR-CIR-SC that combines contrastive learning principles from the SimCLR architecture with spectral clustering (SC) techniques for enhanced feature extraction from channel impulse response (CIR) measurements. Building on the results of autonomous classification, we develop a temporal convolutional network with attention (TCN-A) architecture to discriminate between diverse channel state categories. For each identified channel condition, a dedicated TCN-A model is deployed to predict ranging errors, which subsequently perform distance calibration through error compensation and adaptive weight assignment in the weighted least-square (WLS) positioning algorithm. Experimental evaluations reveal that the proposed SimCLR-CIR-SC method achieves superior autonomous channel state classification and labeling performance compared to three clustering algorithms. Notably, the TCN-A classification model attains an accuracy of 98.16%, surpassing five existing classification models. Furthermore, the proposed positioning method achieves an average error of 0.57 m with three anchors, enhancing the positioning accuracy by at least 31.3% compared to four benchmark methods. Obviously, the positioning accuracy can be further improved as the number of anchors increases, and the average error is 0.278 m with seven anchors.**



***Index Terms*— Channel state, SimCLR-CIR-SC, temporal convolutional network with attention (TCN-A), ultrawideband (UWB), weighted least square (WLS).**

I. Introduction

DRIVEN by the relentless pace of technological and economic progress, the demand for location-based services (LBSs) [1] is growing daily. High-accuracy indoor positioning is increasingly required in various indoor scenarios, such as large shopping centers, underground parking facilities, and comprehensive hospitals. Statistics reveal that individuals spend 80% of their activity time indoors [2], thereby propelling indoor positioning technology into a highly active research area.

Ultrawideband (UWB) technology is regarded as a powerful tool for achieving high-precision indoor positioning due to its high temporal resolution and strong penetration capabilities [3], [4]. The high temporal resolution of UWB signals aids in accurately distinguishing multiple echoes caused by multipath effects, theoretically propelling the positioning accuracy down to the centimeter or even millimeter level [5]. However, in practical scenarios, non-line-of-sight (NLoS) propagation poses a huge challenge, significantly distorting the accuracy of distance measurements between nodes, thereby undermining the precision of range-based UWB positioning systems [6]. Consequently, mitigating range errors caused by NLoS propagation is a critical area of research.

Received 17 January 2025; revised 3 June 2025; accepted 9 June 2025. Date of publication 19 June 2025; date of current version 1 August 2025. This work was supported in part by the Natural Science Foundation of Chongqing under Grant CSTB2024NSCQ-MSX0275. The work of Zheng Chu was supported in part by Ningbo Natural Science Foundation under Grant 2024J233. The associate editor coordinating the review of this article and approving it for publication was Dr. Om Jee Pandey. *(Corresponding author: Xiao Wang.)*

Shixun Wu, Xiao Wang, Miao Zhang, Zhangli Lan, Kai Xu, and Shuang Jin are with the School of Information Science and Engineering, Chongqing Jiaotong University, Chongqing 400074, China (e-mail: wushixun333@163.com; wangxiao5656@126.com; msczz@foxmail.com; 385551137@qq.com; xkxjxwx@hotmail.com; 884580930@qq.com).

Zheng Chu is with the Department of Electrical and Electronic Engineering, University of Nottingham Ningbo China, Ningbo 315100, China (e-mail: andrew.chuzheng7@gmail.com).

Digital Object Identifier 10.1109/JSEN.2025.3579569

1558-1748 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:29:01 UTC from IEEE Xplore. Restrictions apply.

Indoor UWB positioning methods for mitigating NLoS errors can be broadly categorized into two groups [7]. The first group involves the identification and rejection of NLoS. Each transmission link is categorized as either line-of-sight (LoS) or NLoS, where inaccurate range measurement emanating from NLoS is disregarded, and solely the precise range measurement stemming from LoS is employed for position estimation. Traditional methods for NLoS identification include signal power gain utilizing thresholds [8] and constructing LoS/NLoS maps utilizing prior information [9]. Recently, machine learning techniques such as random forest (RF) [10], support vector machines (SVMs) [11], and multilayer perceptions (MLPs) [12] have been extensively applied for NLoS identification. Notably, Sung et al. [13] utilized denoised channel impulse response (CIR) data combined with convolutional neural networks (CNNs) to classify LoS and NLoS, achieving higher classification accuracy than SVM. Furthermore, Fontaine et al. [14] introduced a transfer learning-based framework that not only significantly enhances NLoS identification accuracy through automated hyperparameter tuning but also exhibits superior generalization capabilities across various environments compared to conventional deep learning methods.

The second group is NLoS mitigation and compensation, employing filtering techniques such as Kalman filtering (KF) [15], extended KF (EKF) [16], and particle filtering (PF) [17] to directly diminish the detrimental effects of NLoS errors on position estimation. Additionally, an alternative approach involves adjusting ranging errors and subsequently utilizing these corrected measurements for position estimation. Zhu et al. [18] developed an adaptive error map to compensate for ranging errors under NLoS conditions, effectively enhancing the performance and reliability of the positioning system in a stable static environment. However, its efficacy diminished in a dynamic environment. Yang et al. [19] constructed the error correction-based deep learning (DLRC) network, employing a self-supervised framework to calibrate measured distances, thereby enhancing the accuracy of indoor UWB positioning systems while drastically reducing deployment complexities and costs. Nevertheless, the stability and reliability of this approach necessitated further exploration in fluctuated environments. He et al. [20] proposed a ranging error compensation algorithm that integrates attention mechanisms with deep neural networks (DNNs), significantly reducing ranging errors. This model also exhibited enhanced adaptability and robustness to various environmental changes.

To further mitigate the impact of NLoS errors, it is effective to combine NLoS identification with NLoS compensation. Feng et al. [21] first identified NLoS by SVM based on channel statistics and then adopted EKF to perform NLoS mitigation. However, the NLoS conditions detected with supervised learning models needed labeled training sets, which is a complex and time-consuming task. Kim et al. [22] addressed the issue of labeling training set, proposing a decile labeling strategy for UWB channels based on two-way ranging (TWR) errors, effectively reducing the cost of labeling dataset. Given the complexity and variability of indoor environments, merely categorizing channel states into LoS and NLoS is no longer sufficient for ensuring precision. Although the work in [22] provided a more detailed classification of channel states, ten categories were employed based on empirical grounds, lacking sufficient theoretical justification and explanation. Furthermore, traditional ranging error prediction methods trained the model on entire datasets [19], [20], overlooking data heterogeneity and resulting in limited generalization capabilities, making it difficult to achieve optimal performance in dynamic indoor environments. This study addresses these issues by utilizing unsupervised learning methods to classify channel states based on the inherent characteristics of CIR data. Additionally, multiclass classification and ranging error regression models for each category are constructed. The main contributions of this article are as follows.

1) This research breaks the limitations of traditional LoS/NLoS channel state classification about UWB signal and proposes a SimCLR-CIR-SC method to effectively and autonomously classify the channel states. This method integrates the SimCLR contrastive learning framework with CIR signal characteristics to achieve dynamic autonomous classification through the combination of spectral clustering (SC), a feature gap heuristic algorithm, and a microcluster merging algorithm.
2) Based on the results of SimCLR-CIR-SC method, the temporal convolutional neural network (TCN-A) models with attention mechanism are designed and developed to determine channel states and predict ranging errors, respectively. A new positioning method is proposed through the compensation of range measurements and combination with the weighted least-square (WLS) algorithm, significantly enhancing the positioning accuracy.
3) The experimental results indicate that our proposed SimCLR-CIR-SC method can effectively categorize and label channel states in comparison to three benchmark classification methods. Moreover, our proposed positioning method can accurately determine channel states and estimate range errors. It outperforms four benchmark positioning methods and achieves an average error of 0.57 m with three anchors.

This article is organized as follows. Section II introduces the TCN positioning scheme based on SimCLR-CIR-SC. Section III discusses the SimCLR-CIR-SC autonomous classification method. Sections IV and V detail the model training during the offline stage and the position estimation in the online stage, respectively. Section VI presents the experimental results and analysis. Finally, Section VII concludes this article and discusses future research directions.

II. TCN Positioning Scheme Based on SimCLR-CIR-SC

In UWB systems, the CIR can provide detailed information about the signal propagation in specific environments, including the arrival times, amplitudes, and phases of various multipath components. These characteristics of CIR can reflect the physical and geometrical properties encountered by the signal. Therefore, the CIR facilitates the identification and

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:29:01 UTC from IEEE Xplore. Restrictions apply.

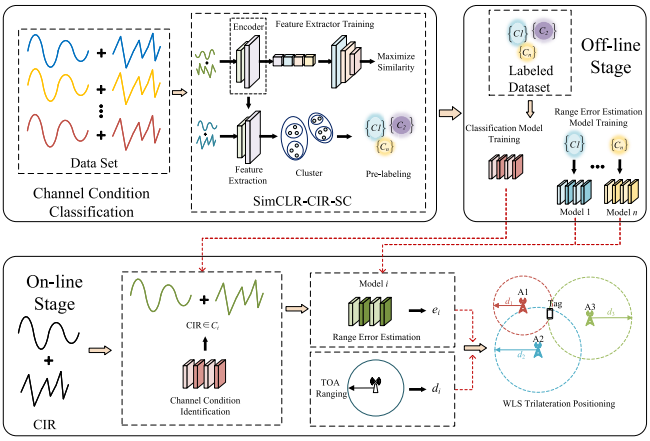


Fig. 1. TCN positioning scheme based on SimCLR-CIR-SC.

distinction of multipath effects, thereby enhancing positioning accuracy [23], [24]. Mathematically, the CIR can be expressed as a weighted sum of impulse functions, where each impulse represents a path and the weights signify the attenuation and phase shifts of that path

$$h(t) = \sum_{n=0}^{N-1} \alpha_n \delta(t - \tau_n) + n(t) \tag{1}$$

where α_n represents the complex gain of the nth path, encompassing amplitude and phase information; τ_n is the propagation delay of the nth path; $\delta(t)$ denotes the Dirac delta function; N is the total number of paths; and $n(t)$ is the additive white Gaussian noise (AWGN) in the channel.

In most cases, channel states are typically categorized into LoS and NLoS, simplifying the identification of channel state as a binary classification problem [25]. However, due to the complexity of indoor environments, the binary classification of LoS and NLoS often fails to accurately represent the varied channel states. Thus, a more detailed and comprehensive classification of channel states is required for a more precise ranging analysis. Furthermore, when supervised learning methods are used to identify channel states, a training set with labeled channel state categories is necessary. Labeling dataset is usually time-consuming and tedious, undoubtedly increasing the implementation cost of the entire positioning system. To address the above issues, this article classifies channel states autonomously based on CIR features, overcoming the inaccuracies and labeling challenges of traditional binary or multiclass channel state descriptions. A TCN positioning scheme based on SimCLR-CIR-SC is proposed, as shown in Fig. 1. The overall positioning scheme is divided into three parts: autonomous classification with SimCLR-CIR-SC, offline model training, and online position prediction.

III. SimCLR-CIR-SC Autonomous Classification

Sensors are capable of capturing multiple consecutive sampling points of CIR data from UWB signals, where a critical segment encompasses the first path with the most useful information. Therefore, a sequence of 152 sampling points started from the first path is selected as the input. Each sampling point contains amplitude and phase information of the CIR, collectively forming a 152 × 2 data matrix. This dual-channel feature selection enhances the model's ability to capture key channel characteristics, improving the accuracy of classification and range error prediction. The resulting 152 × 2 data matrix ensures complete information for subsequent processing and analysis.

The analysis and classification of CIR data inherently relies on sophisticated data mining techniques. Among these, a clustering algorithm is the most commonly used technique, effectively grouping sample data by similarity [26]. Given that the dimension of CIR data is 152 × 2, exhibiting a complex multidimensional structure rather than a simple linear distribution, traditional clustering algorithms may struggle to effectively mine the intrinsic structural properties of CIR data, thus failing to achieve ideal clustering results. In contrast, SC algorithms can detect complex geometric and nonlinear structures, which are capable of processing high-dimensional data and clustering arbitrary shape [27]. Thus, an autonomous classification method termed SimCLR-CIR-SC is proposed, including three parts: SimCLR-CIR feature extraction, feature gap heuristic algorithm, and microcluster merging algorithm.

A. SimCLR-CIR Feature Extraction

As a contrastive learning method, SimCLR provides high-quality features for downstream tasks [28]. However, it is primarily suited for visual image tasks and cannot be directly applied to extract features from time-series CIR data. To address this problem, a SimCLR-CIR feature extraction model specifically tailored for CIR data is devised to extract more refined features. The overall framework, as shown in Fig. 2, includes four key components such as data augmentation, SimCLR-CIR encoder, SimCLR-CIR projection head, and contrastive loss function.

1) CIR Data Augmentation: Data augmentation within the SimCLR framework typically employs standard operations such as image cropping, rotation, and color jittering [29]. For time-series data, detailed studies on SimCLR optimization, including data augmentation, have been conducted [30]. However, this strategy of data augmentation only targets a single dimension, namely amplitude dimension. Our study extends to time-series CIR data, which possess dual dimensions: amplitude and phase. The amplitude represents signal intensity and attenuation, while the phase encodes crucial details about the propagation delays and multipath effects. The utilization of distinct augmentation strategies across both amplitude and phase dimensions substantially enhances the model's adaptability to signal variations in the physical environment.

For the amplitude dimension, random left-to-right flipping has proven to be the most effective method [30]. Accordingly, the random noise injection and random scaling techniques proposed in the same study are also applied. Furthermore, a targeted random smoothing operation is incorporated based on the characteristics of the CIR magnitude, which is utilized

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:29:01 UTC from IEEE Xplore. Restrictions apply.

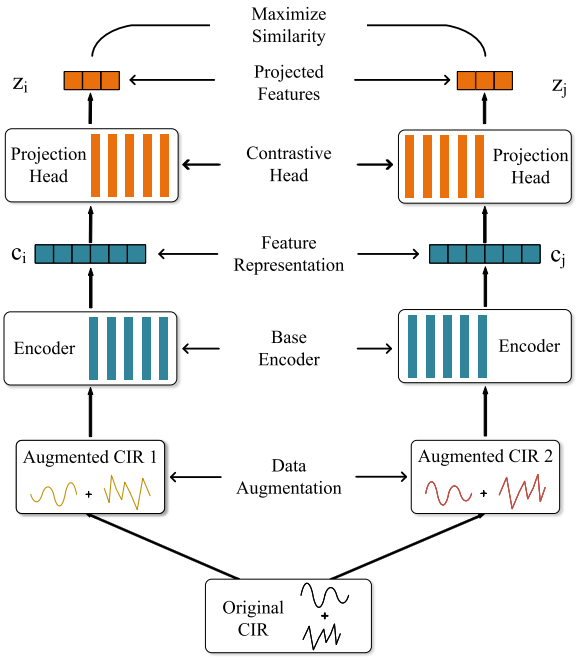


Fig. 2. SimCLR-CIR framework.

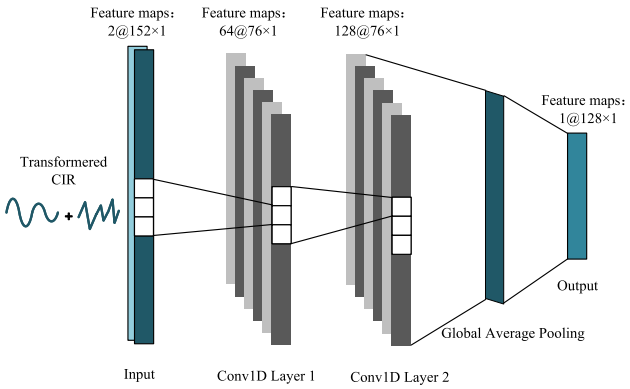


Fig. 3. Encoder framework.

to maintain data integrity and reflect the trends of variability. Smoothing is achieved using an average filter, where the filter size s is randomly chosen from odd values within the range [1, 11], ensuring symmetry at the current point. A filter size $s = 1$ implies the absence of smoothing, maintaining the original data characteristics. Assume that the amplitude of the original signal is m, and the operation can be expressed as follows:

$$m'[i] = \frac{1}{s} \sum_{j=i-\frac{s-1}{2}}^{i+\frac{s-1}{2}} m[j] \tag{2}$$

where $m'[i]$ denotes the smoothed amplitude at the ith point in a CIR amplitude sequence, with the index j ranging from $i - (s - 1/2)$ to $i + (s - 1/2)$. This implies that the smoothed value at each point i is essentially the average of the amplitudes of itself and its neighboring points.

For the phase dimension, random left-to-right flipping is also employed, following the same principle as the amplitude dimension. Additionally, random nonlinear phase transformations and random shifts are introduced. A sinusoidal waveform variation is added to the original phase ϕ, with its amplitude and frequency controlled by random parameters to achieve random nonlinear transformation

$$\phi' = \phi + \beta \cdot \sin(2\pi f t + \theta) \tag{3}$$

where β represents the amplitude, f is the frequency, t denotes the time series, and θ is a random phase shift.

A random shift is added to the phase after nonlinear transformations to simulate the effects of environmental noise and equipment errors

$$\phi'' = \phi' + \delta \tag{4}$$

where δ is the displacement value, uniformly sampled from the interval $[-0.1, 0.1]$.

2) SimCLR-CIR Encoder: Within the SimCLR framework, the encoder functions as a pivotal feature extractor, transforming input images into representations that facilitate subsequent contrastive learning. This process is optimized through the contrastive loss, aiming to maximize the similarity between augmented views, thereby enhancing the ability of model to discern diverse forms of data augmentation [31]. Traditionally, SimCLR utilizes ResNet-50 as the encoder. However, given the nonspatial nature of CIR data, adopting a 1D-CNN model as the encoder for SimCLR-CIR is highly appropriate. Studies have indicated that the 1D-CNN model is exceptionally effective in analyzing time-series data [30], flexibly capturing temporal dynamics within CIR data. This model ensures that the architecture is well-suited to the characteristics of data, enabling more accurate learning and extraction of pertinent information from CIR data.

The encoder architecture, as illustrated in Fig. 3, initiates with a 1D-CNN layer featuring 64 filters, a kernel size of 3, and a stride of 2, activated by ReLU to capture primary time-series features. This layer is followed by batch normalization to stabilize the training process and enhance model generalization. Subsequently, a convolutional layer with 128 filters, retaining a kernel size of 3 and a stride of 1, serves to deepen feature extraction. Finally, a global average pooling layer streamlines the model by reducing parameters and feature dimensionality. This step prepares a condensed and informative feature representation for downstream tasks. Once trained, the encoder is frozen, serving as a dedicated feature extractor within the SimCLR-CIR framework for subsequent tasks.

3) SimCLR-CIR Projection Head: Within the SimCLR framework, the projection head is tasked with dimensionality reduction of the feature vectors extracted by the encoder, helping to filter out potentially irrelevant task-specific details while preserving universally useful features [28]. In contrast to standard SimCLR, SimCLR-CIR innovatively employs a dual-layer fully connected network within its projection head design. The initial layer receives feature vectors from the encoder and undertakes initial nonlinear transformations to enrich the expressiveness of model, providing a more

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:29:01 UTC from IEEE Xplore. Restrictions apply.

TABLE I
PARAMETERS OF THE PROJECTION HEAD

Layer	Number of Units	Activation Function
First Layer	64	Relu
Second Layer	32	None

detailed representation of the input data. The subsequent layer further compacts and optimizes these feature representations, rendering them more suitable for contrastive loss. Details of this projection head configuration are shown in Table I.

4) Contrastive Loss Function: The normalized temperature-scaled cross-entropy loss (NT-Xent) is used as the contrastive loss function for SimCLR-CIR [31]

$$L_{i,j} = -\log \frac{\exp\left(\frac{\text{sim}(z_i, z_j)}{\tau}\right)}{\sum_{k=1}^{2N} \mathbf{1}_{k \neq i} \exp\left(\frac{\text{sim}(z_i, z_k)}{\tau}\right)} \tag{5}$$

where $\text{sim}(z_i, z_k)$ represents the cosine similarity between vectors z_i and z_j, and τ is the temperature parameter. The indicator function $\mathbf{1}_{k \neq i}$, whose value is 1 when $k \neq i$ and 0 otherwise, ensuring that samples are not compared to themselves. This NT-Xent function utilizes the temperature-scaled cross-entropy loss to measure and optimize the similarity between sample pairs, aiming to bring closer the positive pairs from the same sample while pushing apart negative pairs from different samples. Thus, by minimizing the NT-Xent loss, SimCLR-CIR effectively trains the model to learn distinctive and robust representations of CIR signal features.

B. Feature Gap Heuristic Algorithm

The feature gap heuristic algorithm is a method based on spectral graph theory that employs the eigenvalues of the data's Laplacian matrix to estimate the optimal number of clusters. The initial step involves constructing a sparse similarity matrix S with the k-nearest neighbors method. This step efficiently decreases the computational load and storage requirements while retaining crucial local structural information in the dataset. The similarity matrix S is mathematically expressed as follows:

$$S = \begin{cases} 1 - d(i, j), & \text{if } j \in \text{Nearest Neighbors}(i) \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

Constructing the sparse similarity matrix, the cosine distance $d(i, j)$ is used to assess the distance between data points. For each point i, the n nearest neighbors j are identified, and the similarity matrix S is the corresponding entries with $1 - d(i, j)$. This ensures that the values of similarity directly reflect the proximity of the data points, i.e., shorter distances correlate with higher similarities. To ensure the symmetry of the matrix, a further step is required

$$S = \frac{S + S^{\text{T}}}{2}. \tag{7}$$

Subsequently, the normalized Laplacian matrix L_{sym} is derived from the similarity matrix S

$$L_{\text{sym}} = I - D^{-\frac{1}{2}} \text{SD}^{-\frac{1}{2}} \tag{8}$$

where I is the identify matrix and D is a diagonal matrix in which each diagonal element D_{ii} is the sum of the elements in the ith row of the matrix S. The eigenvalues of L_{sym} are calculated as follows:

$$L_{\text{sym}} v = \lambda v \tag{9}$$

where v represents the eigenvector and λ denotes the eigenvalue.

Once all eigenvalues are computed, they are arranged in ascending order. Subsequently, the eigengaps, which are the differences between consecutive eigenvalues, are determined

$$\Delta\lambda_i = \lambda_{i+1} - \lambda_i. \tag{10}$$

The optimal number of clusters, k, is typically selected at the position of the largest eigengap

$$k = \arg\max_i (\Delta\lambda_i) + 1. \tag{11}$$

The rationality of the feature gap heuristic algorithm lies in its ability to infer the inherent clustering structure of data by leveraging the spectral properties of graphs constructed from the data. This algorithm does not rely on any specific assumptions about the data distribution, making it suitable for a wide range of data types and sizes, particularly those with complex, nonlinearly separable structures. Consequently, the feature gap heuristic algorithm offers an intuitive and effective means for determining the number of clusters, providing significant practical value for unsupervised learning and data exploration. Additionally, the construction of a sparse similarity matrix alleviates the computational and storage costs, rendering the algorithm being accurate and practical, especially suitable for dealing with complex datasets in practice.

C. Microcluster Merging Algorithm

The scarcity of adequate sample sizes within cluster categories not only limits the ability of model to learn comprehensive features of the channel but also prevents the model from effectively describing the characteristics of the entire channel environment. Consequently, it is paramount to ensure that each cluster encompasses sufficient samples, enabling a comprehensive capture and reflection of full channel information, thereby enhancing the training performance and model generalization. In this study, in order to improve the stability and sample balance about the outcomes of SC, a microcluster merging algorithm based on cluster centroid distances has been employed. Microcluster is defined as a cluster that the number of samples is fewer than a threshold N_{micro}. Due to the sparsity of data or outliers, microcluster often exists, leading to instability in clustering outcomes. To address this issue, the following strategy is employed for cluster merging.

The first step involves calculating the centroid of each cluster, which serves as the mean vector of all feature vectors within the cluster. Mathematically, for a cluster labeled C_i, its centroid μ_i is defined as

$$\mu_i = \frac{1}{|C_i|} \sum_{l \in C_i} l \tag{12}$$

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:29:01 UTC from IEEE Xplore. Restrictions apply.

where $|C_i|$ represents the number of samples within cluster C_i and l denotes the feature vector of samples in cluster C_i.

The second step involves constructing a distance matrix. The distance matrix, denoted as E, is obtained by calculating the Euclidean distance between every pair of cluster centroids

$$E_{i,j} = ||\mu_i - \mu_j|| \tag{13}$$

where μ_i and μ_j represent the centroids of clusters C_i and C_j, respectively.

The third step involves identifying and merging microclusters. For each microcluster C_k, the nearest major cluster C_j is identified with the distance matrix E, where the size of C_j is greater than or equal to N_{micro}. All samples in C_k are then relabeled as belonging to C_j

$$\text{labels}[c \in C_k] = j. \tag{14}$$

This step is accomplished by identifying the minimum in the distance matrix $E_{i,j}$ without the case when k is equal to j and updating the corresponding cluster labels.

In summary, SimCLR-CIR-SC first performs feature extraction and uses the K-nearest neighbor method to construct a sparse similarity matrix for replacing full density similarity matrix, reducing the computational burden and improving the dimensionality reduction ability of SC. Second, the feature gap heuristic algorithm is used to determine the most suitable number of clusters. Finally, a microcluster merging algorithm ensures the balance among the samples in the clustering results. Algorithm 1 shows the steps of SimCLR-CIR-SC autonomous classification.

Algorithm 1 SimCLR-CIR-SC Autonomous Classification

Input: Dataset X, Number of neighbors n
Output: Clusters $C_1, C_2, \ldots, C_k$

1: Perform feature extraction on dataset X to obtain the feature set.
2: Construct the sparse similarity matrix S using Eq.(6) and Eq.(7).
3: Determine the optimal number of clusters k using Eq.(8), Eq.(9), Eq.(10), and Eq.(11).
4: Perform spectral clustering.
5: Compute the Laplacian matrix L_{sym} using Eq.(8).
6: Perform eigendecomposition on L_{sym} to obtain the eigenvectors as described in Eq.(9).
7: Conduct k-means clustering on the eigenvectors to obtain the clustering labels.
8: Merge smaller clusters into the nearest larger cluster and adjust the clustering labels using Eq.(12), Eq.(13), and Eq.(14).

IV. Offline Model Training

In the offline stage, the SimCLR-CIR-SC autonomous classification method is leveraged to assign labels to various categories, resulting in the generation of a CIR dataset specifically designed for training different models. These models focus on channel state classification and range error prediction. The input of them is 152 × 2 CIR measurements, which include amplitude and phase information of the CIR.

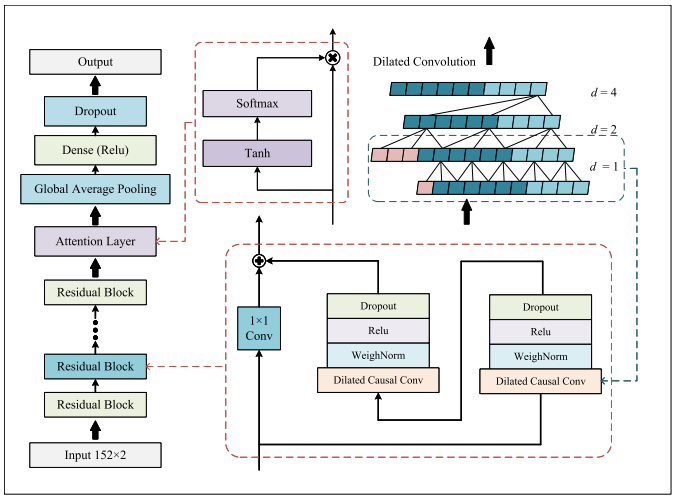


Fig. 4. TCN-A framework. TCN-A architecture employs dilated convolutions and residual connections to capture long-term dependencies and prevent gradient vanishing, enhanced by an attention mechanism for improved temporal pattern recognition and performance.

A. TCN-A Channel State Classification Model

Accurate determination of channel state is crucial for the positioning performance. Incorrect judgments of channel state can directly impact the model selection of range error prediction, potentially leading to the selection of inappropriate models. Inaccuracies predicted by the wrong models can not only fail to reduce positioning errors but also may actually increase them. Therefore, a high-precision channel state classification model is key to ensuring the accuracy of the positioning system.

The TCN structure utilizes dilated convolutions and residual connections to effectively capture long-term dependencies in time-series data, addressing the limitation in traditional CNN where the limited local receptive field impedes the capture of such dependencies. Additionally, residual connections ensure effective information flow even in deeper network layers, overcoming the gradient vanishing problem that is common in CNN. In comparison to LSTM, the parallel processing structure of TCN significantly enhances computational efficiency, particularly when handling large-scale or long sequence data, overcoming the time and resource constraints associated with LSTM's sequential processing. Moreover, TCN avoids the issues of gradient vanishing or exploding that can occur with LSTM in processing long sequences, maintaining the stability of model training. The integration of an attention mechanism at the top of the TCN structure further improves the model's sensitivity to critical temporal features, thereby enhancing the performance in classification and regression tasks. Thus, a TCN-A model is devised to identify the channel state and the structure of TCN-A is illustrated in Fig. 4. The essence of the TCN lies in dilated convolutions to capture long-term dependencies, increasing learning capabilities and network depth by residual connections. Dilated convolutions expand the receptive field of network exponentially without increasing the number of parameters or computational complexity. For 1-D time-series data z, the output $o[t]$ from a dilated convolution can be

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:29:01 UTC from IEEE Xplore. Restrictions apply.

expressed as

$$o[t] = \sum_{k=0}^{K-1} h[k] \cdot z[t - p \cdot k] \tag{15}$$

where h represents the convolutional kernel; K denotes the kernel size; and p is the dilation rate, which determines the interval between elements in the input data. As the network depth increases, the dilation rate p typically grows exponentially (1, 2, 4, 8, ...), rapidly expanding the receptive field.

To prevent the issue of gradient vanishing in deep networks, the TCN incorporates residual connections by constructing residual blocks instead of single dilated convolution layers. The output of each residual block not only includes the results processed through nonlinear activation functions and convolution layers but also the input itself, thereby creating a shortcut path of information

$$z' = \text{Activation}(z + \text{Conv}(z)) \tag{16}$$

where z represents the input and Conv() denotes the output processed through a series of convolutional layers and nonlinear activation functions. If the dimensions of the input and output do not match, a 1×1 convolution can be used to adjust the dimensions.

At the last residual block of TCN-A network, an attention layer is introduced, allowing not only for the capture of long-term dependencies in time-series data but also for dynamically and intelligently focusing on the critical information about the classification or regression tasks. The attention scores at each time step are computed via a linear transformation comprising a weight matrix $\boldsymbol{W}$ and a bias b followed by a nonlinear activation function tanh. These scores are then normalized using the Softmax function to obtain the final attention weights, which are multiplied by the input to produce the weighted feature output. The specific formula is as follows:

$$a = \tanh(z'\boldsymbol{W} + b) \tag{17}$$

$$\alpha = \text{Softmax}(a) \tag{18}$$

$$z'' = z' \odot \alpha. \tag{19}$$

Following the attention layer is a global average pooling layer that reduces the temporal dimension, simplifying the output of network and providing a more refined feature representation for classification or regression tasks. Subsequently, a fully connected layer with ReLU activation function further integrates the features and includes intermediate layers for additional nonlinear processing. A dropout layer follows the fully connected layer to further prevent overfitting. Finally, the output layer uses the Softmax function to transform the output of network into a probability distribution. The probability of each category corresponds to a type of channel state.

B. TCN-A Range Error Prediction Model

The range error prediction model also employs the TCN-A network. Although the overall architecture is consistent with channel state classification model, the output layer is specifically designed with a single neuron to directly predict a numerical value. Additionally, we have tuned the pivotal hyperparameters of the model, encompassing dilation rate, the quantity of filters within residual blocks, kernel size, and the loss function. These adjustments are aimed at precisely tailoring the model to meet specific demand for range measurement, thereby enhancing the accuracy of predictions.

V. Online Position Prediction

In the online stage, the channel state classification model initially determines the state of channel using real-time CIR data of UWB signal. Subsequently, based on the outcome of classification, an appropriate range error prediction model is selected to estimate the range errors. These errors are then utilized to compensate the measured distance, thereby completing the position prediction with the WLS positioning algorithm.

Assume that the coordinate of a tag in a 2-D plane is (x, y) and the coordinates of the anchors are (x_i, y_i) for $i = 1, \ldots, N$. Considering the potential errors, the distance measurement r_i from the tag to each anchor is calculated as

$$r_i = d_i + e_i = \sqrt{(x - x_i)^2 + (y - y_i)^2} + e_i \tag{20}$$

where d_i is the actual distance and e_i represents the ranging error, which may be caused by factors such as NLoS propagation, accuracy of device, and other environmental variables.

The estimated range error $\hat{e}_i$, obtained from the position prediction model, is used to compensate the measured distance, resulting in an adjusted range estimate $\hat{d}_i$ that is closer to the true distance d_i. The formula is as follows:

$$\hat{d}_i = d_i - \hat{e}_i. \tag{21}$$

By defining a new variable $R = x^2 + y^2$ and constructing the unknown vector $\boldsymbol{\theta}_{\text{WLS}} = [x, y, R]^{\text{T}}$, the nonlinear objective function of (23) can be transformed into a linear form [33]. Then, the position of tag is solved using the WLS algorithm as follows:

$$\hat{\boldsymbol{\theta}}_{\text{WLS}} = (\mathbf{H}^{\mathbf{T}}\boldsymbol{W}\mathbf{H})^{-1}\mathbf{H}^{\mathbf{T}}\boldsymbol{W}\mathbf{B} \tag{22}$$

where

$$\mathbf{H} = \begin{bmatrix} -2x_1 & -2y_1 & 1 \\ -2x_2 & -2y_2 & 1 \\ \vdots & \vdots & \vdots \\ -2x_N & -2y_N & 1 \end{bmatrix} \tag{22a}$$

$$\mathbf{B} = \begin{bmatrix} \hat{d}_1^2 - x_1^2 - y_1^2 \\ \hat{d}_2^2 - x_2^2 - y_2^2 \\ \vdots \\ \hat{d}_N^2 - x_N^2 - y_N^2 \end{bmatrix} \tag{22b}$$

$$\boldsymbol{W} = \begin{bmatrix} w_1 & 0 & \ldots & 0 \\ 0 & w_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & w_N \end{bmatrix}. \tag{22c}$$

Each weight w_i corresponds to the weight of an individual range measurement. The estimated range errors precisely

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:29:01 UTC from IEEE Xplore. Restrictions apply.

TABLE II
PART ITEMS OF THE DATASET

Column Names	Description
tag_id	Tag ID
anchor_id	Anchor ID
x_tag	Tag position on x-axis
y_tag	Tag position on y-axis
x_anchor	Anchor position on x-axis
y_anchor	Anchor position on y-axis
NLOS	NLoS situation: 1 if NLoS, otherwise 0
range	Measured range
fp_index	Index of the first path in the CIR
CIR	List of CIR values in a complex form

TABLE III
RESULTS OF CLASSIFICATION ALGORITHM WITH VARYING CLUSTER NUMBERS

Category Number Determination Method	Number of Categories	Sample Count per Category	SS	DB
Predetermined	3	Cluster 0: 10282 Cluster 1: 11305 Cluster 2: 21613	0.289	1.09
Predetermined	4	Cluster 0: 17420 Cluster 1: 6131 Cluster 2: 3750 Cluster 3: 15899	0.241	1.33
Predetermined	5	Cluster 0: 11376 Cluster 1: 13640 Cluster 2: 3821 Cluster 3: 2392 Cluster 4: 11971	0.204	1.31
Feature Gap Heuristic and Micro-Cluster Merging	4	Cluster 0: 10430 Cluster 1: 11648 Cluster 2: 10714 Cluster 3: 10408	0.270	1.06

reflect the expected accuracy of the current range measurements, a smaller estimated range error indicates higher expected precision, and vice versa. Therefore, the weight can be optimized based on the estimated range error $\hat{e}_i$. The formula for each weight w_i in the weight matrix $\boldsymbol{W}$ is as follows:

$$w_i = \frac{1}{\hat{e}_i^2 + \varepsilon} \tag{23}$$

where ε is a small constant introduced to prevent the weight w_i to be zero. This ensures that range measurements with smaller range errors receive higher weights, making the weight matrix $\boldsymbol{W}$ more rational and effectively improving the reliability of the positioning solution.

VI. EXPERIMENTAL RESULTS AND ANALYSIS

This section initially introduces the dataset employed to assess the efficacy of various methods, and then, three interconnected and mutually complementary components of the experimental design are presented. First, the performance of autonomous channel state classification utilizing the SimCLR-CIR-SC method is evaluated. Subsequently, the results from this classification are utilized to analyze the TCN-A-based channel state recognition model and the range error estimation model. Finally, the trained models are integrated into a WLS-based positioning scheme for error compensation, and the positioning performance is validated through comparative experiments.

A. Experimental Dataset

This study employs a publicly available dataset released in Scientific Data in 2023 [34] for performance comparison and analysis. The dataset was meticulously collected in four different rooms equipped with eight anchors and one tag, effectively simulating human walking paths. Sampling was conducted at uniformly spaced intervals, collecting various distance measurements and CIR data for each position of tag, as illustrated in Table II. For this experiment, samples from environment 0 and environment 2 were extracted to produce a comprehensive dataset comprising a total of 43 200 data points.

B. Experimental Results of Autonomous Classification

To facilitate intuitive interpretation of the clustering outcomes, dimensionality reduction is achieved through principal component analysis (PCA), enabling visualization on a 2-D scatter plot. These plots provide a preliminary visual assessment of clustering quality. Ideally, clusters should be simple and consistent in shape without any unusual extensions or anomalies and possess clear cluster boundaries. Additionally, the size of cluster should be balanced to ensure that no single cluster significantly outweighed others, maintaining sample uniformity.

1) Evaluation Metric: The silhouette score (SS) is a metric for assessing the efficacy of clustering algorithms [37]. This metric describes both the compactness within clusters and the degree of separation between them. The formula for calculating SS is as follows:

$$\mathrm{SS} = \frac{b' - a'}{\max(a', b')} \tag{24}$$

where a' represents the average distance of each data point to all other points within its own cluster, effectively measuring the degree of a point that is closely surrounded by its cluster. Conversely, b' denotes the average distance from the data point to all the points in the nearest cluster. The SS, ranging from -1 to 1, provides valuable insights into the quality of the clustering. SS close to 1 indicates a highly appropriate and well-defined clustering, whereas SS near 0 may suggest overlapping clusters. In addition, SS approaching to -1 indicates a likely poor performance of the clustering algorithm.

The Davies–Bouldin (DB) index is another metric used to evaluate the quality of clustering algorithm [38]. DB is based on the similarity within clusters and the difference between clusters, and its calculation formula is as follows:

$$\mathrm{DB} = \frac{1}{k}\sum_{i=1}^{k} \max_{j \neq i} \left(\frac{N_i + N_j}{M_{ij}} \right) \tag{25}$$

where N_i represents the average distance from all points within cluster i to the cluster center. For each pair of clusters i, j, M_{ij} is the distance between the centers of these clusters. k is the total number of clusters. For each cluster i, the maximum ratio $\max_{j \neq i}(N_i + N_j / M_{ij})$ is found, and DB is

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:29:01 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
METRICS OF LABELING RESULTS

Method	Number of Categories	Sample Count per Category	SS	DB
K-Means [35]	4	Cluster 0: 15473 Cluster 1: 11444 Cluster 2: 7042 Cluster 3: 9241	0.061	3.44
Fuzzy C-Means [36]	4	Cluster 0: 12188 Cluster 1: 10201 Cluster 2: 9596 Cluster 3: 11215	0.065	4.05
Original SC [27]	4	Cluster 0: 18114 Cluster 1: 24960 Cluster 2: 60 Cluster 3: 66	0.048	2.13
SimCLR-CIR-SC	4	Cluster 0: 10430 Cluster 1: 11648 Cluster 2: 10714 Cluster 3: 10408	0.270	1.06

the average of these ratios for all clusters. A lower DB indicates higher clustering quality, implying greater separation and compactness among the clusters.

2) Results of Classification Category Counts: This section aims to validate the effectiveness of feature gap heuristic and microcluster merging algorithms. Specifically, the study compares the cluster numbers determined by feature gap heuristic and microcluster merging algorithms with predetermined cluster numbers of 3, 4, and 5 for evaluation. According to the results presented in Table III, the predetermined category number of three outperforms the predetermined category numbers of 4 and 5 in terms of both SS and DB. However, when considering both clustering performance and sample distribution balance, the feature gap heuristic combined with the microcluster merging algorithm achieves the optimal balance in a four-category classification scenario. This approach not only maintains good clustering performance but also ensures a balanced distribution of samples. Such results provide a reliable foundation for the robust training of subsequent channel recognition models.

3) Classified Results and Analysis: In order to validate the performance of the proposed SimCLR-CIR-SC method, where a comparative analysis is conducted using three benchmark methods: K-means [35], fuzzy C-means [36], and the original SC [27]. Each of these methods is configured to discern four distinct clusters, ensuring a consistent basis for comparison. Notably, the original SC utilizes the same sparse similarity matrix as the SimCLR-CIR-SC method, which involved a neighborhood size of $n = 50$.

The experimental results of four clustering methods are depicted in Fig. 5. Fig. 5(d) shows the labeled dataset using the SimCLR-CIR-SC method, which clearly demonstrates the most coherent clustering results. Each cluster displays a high degree of aggregation with distinct boundaries and minimal overlap between clusters. The shape and distribution are relatively uniform. Fig. 5(a) and (b) presents the scatter plots of the labeled datasets using K-means and fuzzy C-means methods, respectively. Although these charts demonstrate a good clustering effect, there are still noticeable overlaps between some clusters, with blurred boundaries. Fig. 5(c) illustrates the labeled dataset using the original SC method, which fails to effectively cluster the data. Most data points are concentrated in two large clusters, while the boundaries of other smaller clusters intersect with those of the large ones, indicating that the original SC method struggles to cluster CIR data with complex and dense internal structures.

To further compare the performance of four clustering methods, Table IV comprehensively outlines the key metrics, encompassing the number of categories, the quantity of samples per category, as well as the calculated SS and DB. The results presented in Table IV indicate that the SimCLR-CIR-SC method outperforms K-means, fuzzy C-means, and original SC in effectively clustering CIR data, ensuring a balanced distribution of samples among clusters. Notably, the SimCLR-CIR-SC method exhibits remarkable performance enhancements in terms of SS and DB compared to the original SC. Despite SS whose value is 0.27 not approaching one, its positivity signifies that a considerable proportion of samples are adequately assigned to their respective clusters. This also indicates a degree of cohesion and separation within the clustering, though there is potential for further refinement. Furthermore, the relatively low DB whose value is 1.06 signifies the high quality of clustering results, characterized with compact clusters that are appropriately distanced from each other.

C. Experimental Results of TCN-A Model Training

1) Parameters Setting and Evaluation Metric: After the CIR dataset has been labeled with channel states, 70% of it is randomly selected as the training set, while the remaining 30% serves as the test set. Subsequently, the TCN-A network is trained to function as both channel state classification model and ranging error prediction model.

For the channel state classification model, performance is evaluated and compared with four key metrics: accuracy, precision, recall, and $F1$-score [39]. The TCN-A network is configured with a dilation rate [1, 2, 4, 8, 16], 32 filters per layer, and a kernel size of 3 and employs the Adam optimizer with a sparse classification cross-entropy loss function.

For the range error prediction model, mean squared error (mse) is used as the performance metric [40]. A lower mse indicates that the model predictions are closer to the actual data, thereby indicating superior model performance. The TCN-A network employs a dilation rate [1, 2, 4, 8, 16, 32], with 64 filters per layer and a kernel size of 5, using the Adam optimizer and mse loss function.

2) Results of Channel State Classification Model: To validate the effectiveness of our proposed TCN-A model for classification, its performance is compared with five existing models: convolutional neural network (CNN) [12], long short-term memory network (LSTM) [41], gated recurrent unit (GRU) [42], bidirectional LSTM (BiLSTM) [43], and TCN [44].

The results presented in Table V indicate that the TCN-A model outperforms all other models in every metric, achieving accuracy and $F1$-scores of 98.16% and 98.17%, respectively. This demonstrates the excellent capability of model to balance

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:29:01 UTC from IEEE Xplore. Restrictions apply.

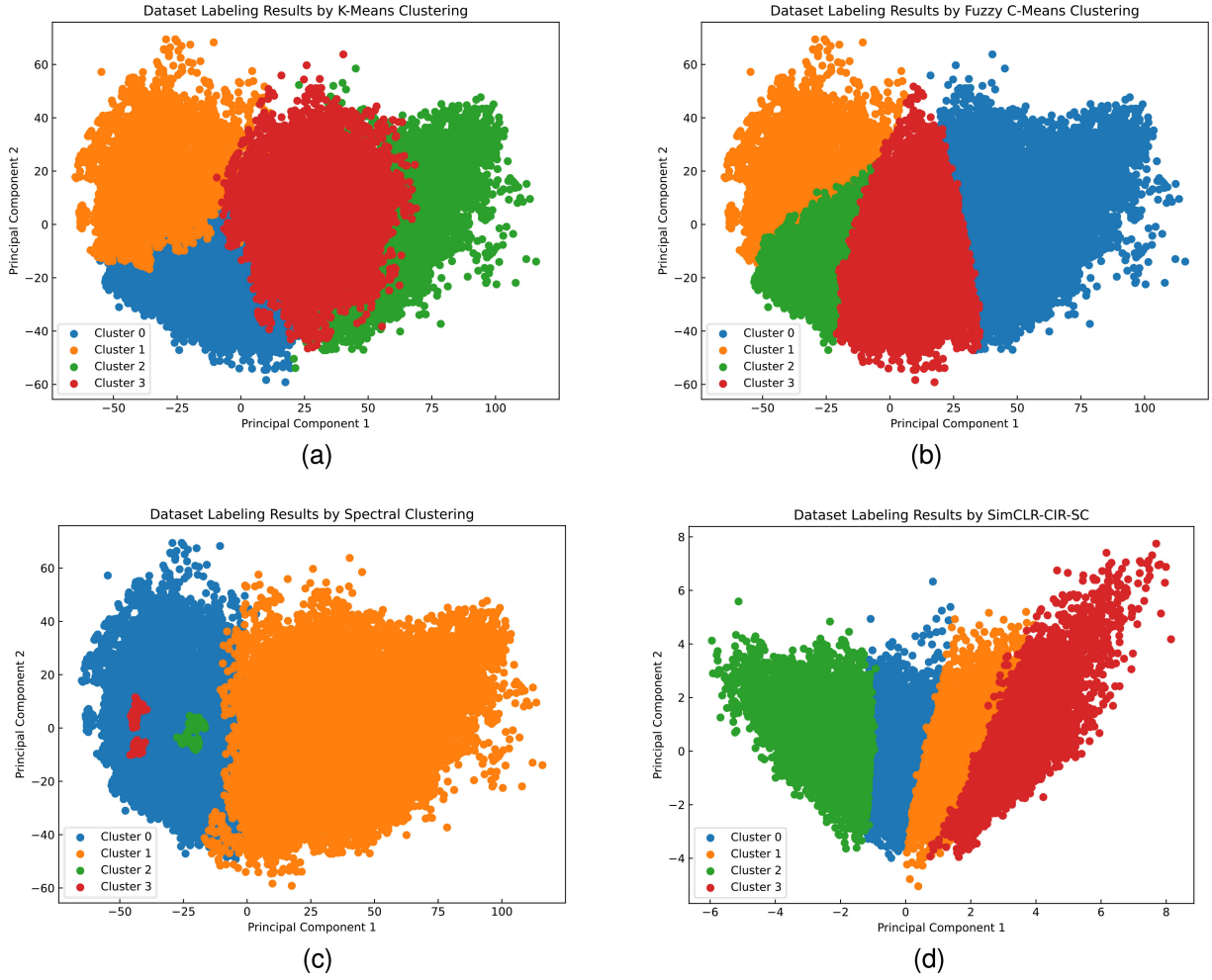


Fig. 5. Scatter plot of labeling results. (a) Scatter plot of *K*-means. (b) Scatter plot of fuzzy C-means. (c) Scatter plot of original SC. (d) Scatter plot of SimCLR-CIR-SC.

TABLE V
PERFORMANCE COMPARISON OF CLASSIFICATION MODEL

Model	Accuracy [%]	Precision [%]	Recall [%]	F1 score [%]
CNN [12]	91.64	91.71	91.71	91.69
LSTM [41]	95.29	95.30	95.34	95.31
GRU [42]	96.12	96.18	96.12	96.14
BiLSTM [43]	96.06	96.11	96.06	96.07
TCN [44]	97.22	97.29	97.21	97.24
TCN-A	**98.16**	**98.16**	**98.18**	**98.17**

TABLE VI
PERFORMANCE COMPARISON OF RANGE ERROR PREDICTION MODEL

Model	MSE
CNN	0.105
TCN-A	0.094
TCN-A(Label00)	0.020
TCN-A(Label01)	0.044
TCN-A(Label02)	0.039
TCN-A(Label03)	0.018

precision and recall, effectively identifying and correctly classifying most of the samples. Despite the existing models, such as CNN, LSTM, GRU, and BiLSTM, also displaying commendable performance, TCN and its enhanced version TCN-A show more significant improvements when dealing with CIR data that exhibit complex temporal dependencies. The unique dilated convolutional structure of TCN provides deeper sequence understanding, likely contributing to its superior performance over BiLSTM structures. The integration of an attention mechanism within the TCN-A model further enhances the sensitivity of model to crucial information by dynamically adjusting its focus, particularly vital in multiclass labeling tasks, thereby achieving higher accuracy and $F1$-scores.

Furthermore, the SimCLR-CIR-SC method effectively clusters similar feature data, enabling the network to learn and capture critical data structures, thereby enhancing the accuracy of channel state recognition. The experimental results show that the CNN model trained on the processed dataset achieves a four-class classification accuracy of 91.64%, whereas the CNN binary classifier trained on the same raw dataset in [26] reports an accuracy of 87.4%. The improved performance of the four-class model, despite the increased task complexity, suggests that the SimCLR-CIR-SC method is beneficial in constructing a more structured and informative feature space. This approach may provide relatively higher quality inputs for downstream classification models, potentially enhancing learning efficiency and recognition accuracy.

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:29:01 UTC from IEEE Xplore. Restrictions apply.

3) Results of Ranging Error Prediction Model: Initially, the range error prediction model is trained with the TCN-A network for each prelabeled category within the dataset. Subsequently, its performance is compared with two models: a CNN-based range error prediction model [26] and the TCN-A model, and both of them are trained directly on the entire dataset without prior labeling.The experimental results are shown in Table VI.

From the results presented in Table VI, it is evident that the TCN-A models trained on prelabeled categories exhibit significant differences in mse performance. Compared to the CNN model and TCN-A model trained on the entire dataset, the TCN-A model trained on specific labeled categories demonstrates superior performance. Specifically, the mse for the TCN-A model trained on the entire dataset is 0.094, but prelabeling substantially reduces the mse of corresponding categories, particularly for "Label00" and "Label03." Their mses drop to 0.02 and 0.018, indicating exceptional accuracy of range error prediction. This improvement indicates that the prelabeling process helps to reveal the intrinsic structure of the CIR data, enabling the model to learn and adapt more effectively to each specific channel state. Furthermore, the performance enhancement of models trained separately for each labeled category confirms the effectiveness of optimizing model parameters for specific classes. Overall, these experimental results underscore the importance of targeted data preprocessing and labeling in enhancing the performance of deep learning models, particularly dealing with complex channel states of UWB signals.

D. Comparison of Positioning Performance

This part of experiment assesses the performance of various positioning methods, and the results are obtained with three anchors. The test utilizes 4000 samples from environments 0 and 2 in the dataset, with each positioning test conducted 800 times. To validate the proposed positioning method, four existing methods are compared and analyzed: the original WLS positioning algorithm, CNN-based NLoS identification WLS (NIDC-WLS) [45], CNN-based error compensation WLS (ECC-WLS) [26], and multiclass LSTM-WLS (MC-LSTM-WLS) [22]. The NIDC-WLS approach is based on the binary classification of LoS/NLoS signals, utilizing a trained channel state identification model to exclude NLoS signals and rely solely on LoS signals for positioning. The ECC-WLS method employs a CNN-based ranging error compensation model to correct the ranging, thereby reducing the positioning errors. The MC-LSTM-WLS scheme combines channel state identification with ranging error compensation, leveraging LSTM networks to recognize channel categories and apply error compensation based on the average ranging error and variance of each channel category.

Positioning error is measured as the Euclidean distance between the actual and estimated coordinates. For each method, the average error, standard deviation of error, and median error are computed after 800 trials; the formulas for the average error and standard deviation of error are as follows:

$$e_{p_c} = \sqrt{(x_c - \hat{x}_c)^2 + (y_c - \hat{y}_c)^2} \tag{26}$$

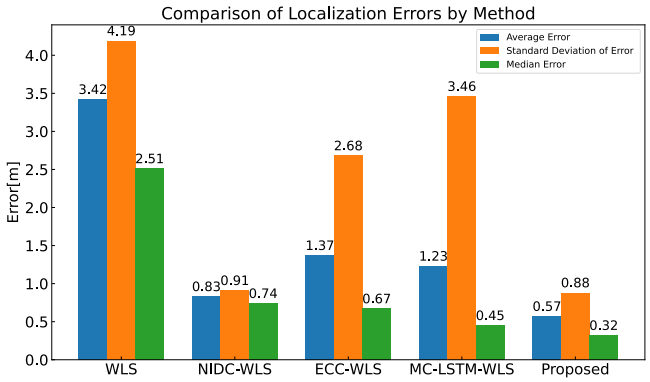


Fig. 6. Comparison of positioning errors.

$$\mu_p = \frac{1}{n}\sum_{c=1}^{n} e_{p_c} \tag{27}$$

$$\sigma_p = \sqrt{\frac{1}{n}\sum_{c=1}^{n}(e_{p_c} - \mu_p)^2} \tag{28}$$

where e_{p_c} denotes the positioning error for the cth trial and (x_c, y_c) and $(\hat{x}_c, \hat{y}_c)$ represent the actual and estimated coordinates of the tag, respectively. The average error, μ_p, reflects the average level of positioning errors across all trials. This provides a crucial metric for assessing the overall accuracy of the positioning system. The standard deviation of error, σ_p, quantifies the variability of these errors relative to the average, offering an intuitive measure of the system's stability.

In Fig. 6, the average error is depicted in blue, the standard deviation of error is depicted in orange, and the median error is depicted in green. Based on these metrics, the proposed method in this study shows the highest precision, with less variability and more stability in the results. Specifically, the proposed method achieves an average error of 0.57 m, making a substantial 31.3% enhancement over the 0.83-m range error of NIDC-WLS. ECC-WLS and MC-LSTM-WLS perform slightly worse, with average errors of 1.37 and 1.23 m, respectively. Conversely, the traditional WLS method, which does not account for the effects of NLoS, exhibits the poorest accuracy with an average error of 3.42 m. These results confirm the severe impact of NLoS on the accuracy of UWB positioning and demonstrate the effectiveness and reliability of our proposed method in complex indoor environments. This improvement is primarily attributed to two aspects: the use of SimCLR-CIR-SC for effective channel state classification, enabling condition-specific error modeling; and the integration of TCN-A-based estimators with WLS compensation, which together improve robustness under NLoS conditions.

Fig. 7 illustrates the comparison of cumulative distribution functions (CDFs) for positioning errors across different methods, and it can be observed that the performance of the proposed method is the best. Specifically, the proposed method rapidly achieves high cdf values within a smaller error range, indicating that most positioning results have minimal error. For instance, the proposed method maintains positioning

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:29:01 UTC from IEEE Xplore. Restrictions apply.

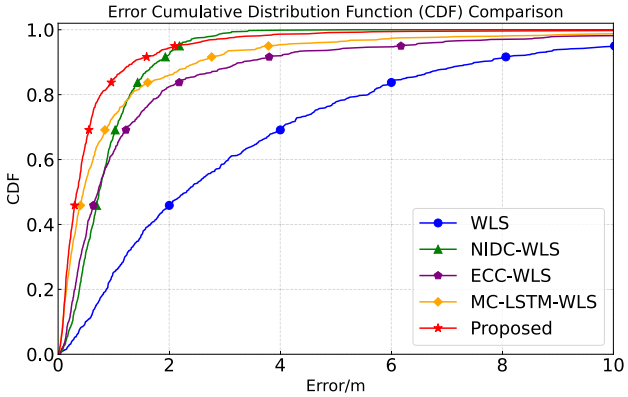


Fig. 7. CDF of positioning error.

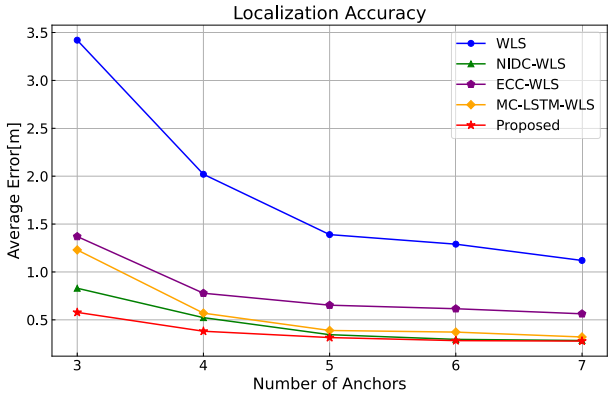


Fig. 8. Positioning accuracy with different numbers of anchors.

errors within 1 m with 85.4% probability, significantly surpassing other methods. The traditional WLS method without processing NLoS exhibits the poorest performance, while the ECC-WLS, which compensates for range errors directly, shows moderate performance. It indicates that the improvements from merely relying on model-based range error compensation without channel state classification are limited. In contrast, NIDLS and MC-LSTM-WLS perform slightly better, suggesting that channel state classification enhances positioning accuracy. However, they still fall behind the proposed method. These results demonstrate that optimizing the model structure and labeling channel state can significantly improve the positioning accuracy, effectively reduce the error distribution, and thus enhance the practicality and reliability of the positioning system.

The experimental results depicted in Fig. 8 demonstrate that the proposed method achieves the highest positioning accuracy with different numbers of anchors. Its average errors decrease steadily from 0.578 m with three anchors to 0.278 m with seven anchors, significantly outperforming other comparative methods. Notably, compared to the conventional WLS approach, the proposed method exhibits superior accuracy even with the minimum anchor configuration and more stable error reduction, highlighting its significant advantages in channel feature extraction and spatial information utilization. This outstanding performance validates the integration of advanced channel state classification strategies with the network model, significantly enhancing the positioning precision. The NIDC-WLS, MC-LSTM-WLS, and the proposed method consistently outperform the other two methods across different numbers of anchors, emphasizing the crucial impact of channel state on indoor UWB positioning performance.

VII. Conclusion and Future Work

This article proposes a novel positioning method based on SimCLR-CIR-SC autonomous channel state classification, which successfully addresses the limitations of traditional classification methods in complex indoor environments. By integrating the SimCLR-CIR-SC autonomous classification method, we not only enhance the precision of channel state descriptions but also significantly improve the robustness and accuracy of the positioning method through data labeling. Our method comprises three key phases: autonomous channel state classification, offline model training, and online position prediction, ensuring the efficiency and practicality of the positioning solution. The experimental results demonstrate that our proposed method substantially outperforms the existing methods in terms of positioning precision and adaptability.

Future work will focus on further optimizing the autonomous channel state classification method, enhancing the SS to achieve better classification performance, and planning data collection to validate the practicality and effectiveness of our proposed method. Additionally, the integration of other sensors, such as inertial measurement units (IMUs), will be explored to improve positioning accuracy.

References

[1] X. Gong, J. Liu, S. Yang, F. Gu, G. Huang, and Y. Bai, "An enhanced indoor positioning solution using dynamic radio fingerprinting spatial context recognition," *IEEE Internet Things J.*, vol. 10, no. 2, pp. 1297–1309, Jan. 2023.

[2] Z. Deng, Y. Yu, X. Yuan, N. Wan, and L. Yang, "Situation and development tendency of indoor positioning," *China Commun.*, vol. 10, no. 3, pp. 42–55, Mar. 2013.

[3] J. Liu, G. Huang, J. Hyyppä, J. Li, X. Gong, and X. Jiang, "A survey on location and motion tracking technologies, methodologies and applications in precision sports," *Expert Syst. Appl.*, vol. 229, Nov. 2023, Art. no. 120492.

[4] G. Yufeng, D. Yujie, W. Yuyang, L. Kunpeng, and L. Chengshan, "Research on ultra-wideband positioning algorithm based on neural networks and self-tuning Kalman filter," *China Mech. Eng.*, vol. 34, no. 12, pp. 1504–1511, 2023.

[5] D. Chiasson, Y. Lin, M. Kok, and P. B. Shull, "Asynchronous hyperbolic UWB source-localization and self-localization for indoor tracking and navigation," *IEEE Internet Things J.*, vol. 10, no. 13, pp. 11655–11668, Jul. 2023.

[6] H. Xu, H. Liu, and H. Shao, "Indoor positioning method based on UWB and weighted concentric circle clustering," *J. Electron. Meas. Instrum.*, vol. 38, no. 4, pp. 161–175, Apr. 2024.

[7] E. Dehmollaian, B. Etzlinger, P. Peterseil, and A. Springer, "AI-enhanced UWB-based localisation in wireless networks," in *Intelligent Secure Trustable Things*, vol. 1147. Cham, Switzerland: Springer, Jun. 2024, pp. 201–216.

[8] W. Sun and P. Sun, "A method for UWB anomaly signal recognition based on machine learning classifiers," *Sci. Surveying Mapping*, vol. 48, no. 5, pp. 1–8+34, May 2023.

[9] Q. Wang, Z. Li, H. Zhang, Y. Yang, and X. Meng, "An indoor UWB NLOS correction positioning method based on anchor LOS/NLOS map," *IEEE Sensors J.*, vol. 23, no. 24, pp. 30739–30750, Dec. 2023.

[10] Y. Yang, L. Zhang, J. Xu, D. Li, J. Bao, and J. Tan, "Cooperative indoor localization system based UWB and random forest algorithm in complicated underground NLOS scenario," in *Proc. 9th Int. Conf. Digit. Home (ICDH)*, Oct. 2022, pp. 271–276.

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:29:01 UTC from IEEE Xplore. Restrictions apply.

[11] H. Yang, Y. Wang, C. K. Seow, M. Sun, M. Si, and L. Huang, "UWB sensor-based indoor LOS/NLOS localization with support vector machine learning," *IEEE Sensors J.*, vol. 23, no. 3, pp. 2988–3004, Feb. 2023.

[12] M. Si, Y. Wang, H. Siljak, C. Seow, and H. Yang, "A lightweight CIR-based CNN with MLP for NLOS/LOS identification in a UWB positioning system," *IEEE Commun. Lett.*, vol. 27, no. 5, pp. 1332–1336, May 2023.

[13] S. Sung, H. Kim, and J.-I. Jung, "Accurate indoor positioning for UWB-based personal devices using deep learning," *IEEE Access*, vol. 11, pp. 20095–20113, 2023.

[14] J. Fontaine et al., "Transfer learning for UWB error correction and (N)LOS classification in multiple environments," *IEEE Internet Things J.*, vol. 11, no. 3, pp. 4085–4101, Feb. 2024.

[15] B. Venkata Krishnaveni, K. Suresh Reddy, and P. Ramana Reddy, "Indoor positioning and tracking by coupling IMU and UWB with the extended Kalman filter," *IETE J. Res.*, vol. 69, no. 10, pp. 6757–6766, Oct. 2023.

[16] G. Guan and G. Chen, "An improved LSE-EKF optimisation algorithm for UAV UWB positioning in complex indoor environments," *J. Control Decis.*, vol. 10, no. 4, pp. 547–559, Oct. 2023.

[17] Y. Li, Z. Gao, Q. Xu, and C. Yang, "Comprehensive evaluations of NLOS and linearization errors on UWB positioning," *Appl. Sci.*, vol. 13, no. 10, p. 6187, May 2023.

[18] X. Zhu, J. Yi, J. Cheng, and L. He, "Adapted error map based mobile robot UWB indoor positioning," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 9, pp. 6336–6350, Sep. 2020.

[19] B. Yang, J. Li, Z. Shao, and H. Zhang, "Self-supervised deep location and ranging error correction for UWB localization," *IEEE Sensors J.*, vol. 23, no. 9, pp. 9549–9559, May 2023.

[20] X. He, L. Mo, and Q. Wang, "An attention-assisted UWB ranging error compensation algorithm," *IEEE Wireless Commun. Lett.*, vol. 12, no. 3, pp. 421–425, Mar. 2023.

[21] D. Feng et al., "An adaptive IMU/UWB fusion method for NLOS indoor positioning and navigation," *IEEE Internet Things J.*, vol. 10, no. 13, pp. 11414–11428, Jul. 2023.

[22] D.-H. Kim, A. Farhad, and J.-Y. Pyun, "UWB positioning system based on LSTM classification with mitigated NLOS effects," *IEEE Internet Things J.*, vol. 10, no. 2, pp. 1822–1835, Jan. 2023.

[23] Z. Li, A. Nimr, P. Schulz, and G. Fettweis, "Superresolution wireless multipath channel path delay estimation for CIR-based localization," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Apr. 2022, pp. 1940–1945.

[24] V. Tran, Z. Dai, N. Trigoni, and A. Markham, "DeepCIR: Insights into CIR-based data-driven UWB error mitigation," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2022, pp. 13300–13307.

[25] X. Yang, J. Wang, D. Song, B. Feng, and H. Ye, "A novel NLOS error compensation method based IMU for UWB indoor positioning system," *IEEE Sensors J.*, vol. 21, no. 9, pp. 11203–11212, May 2021.

[26] K. Bregar and M. Mohorcic, "Improving indoor localization using convolutional neural networks on computationally restricted devices," *IEEE Access*, vol. 6, pp. 17429–17441, 2018.

[27] J. Xie, W. Kong, S. Xia, G. Wang, and X. Gao, "An efficient spectral clustering algorithm based on granular-ball," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 9, pp. 9743–9753, Sep. 2023.

[28] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. 37th Int. Conf. Mach. Learn.*, vol. 119, Jul. 2020, pp. 1597–1607.

[29] J. Cui, Z. Zhong, S. Liu, B. Yu, and J. Jia, "Parametric contrastive learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 695–704.

[30] J. Pöppelbaum, G. S. Chadha, and A. Schwung, "Contrastive learning based self-supervised time-series analysis," *Appl. Soft Comput.*, vol. 117, Mar. 2022, Art. no. 108397.

[31] C. H. Yeh, C. Y. Hong, Y. C. Hsu, T. L. Liu, Y. Chen, and Y. LeCun, "Decoupled contrastive learning," in *Proc. 17th Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Oct. 2022, pp. 668–684.

[32] K.-H. Lee, A. Arnab, S. Guadarrama, J. Canny, and I. Fischer, "Compressive visual representations," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, Dec. 2021, pp. 19538–19552.

[33] J. Wei, H. Wang, S. Su, Y. Tang, X. Guo, and X. Sun, "NLOS identification using parallel deep learning model and time-frequency information in UWB-based positioning system," *Measurement*, vol. 195, May 2022, Art. no. 111191.

[34] K. Bregar, "Indoor UWB positioning and position tracking data set," *Scientific Data*, vol. 10, no. 1, p. 744, Oct. 2023.

[35] A. M. Ikotun, A. E. Ezugwu, L. Abualigah, B. Abuhaija, and J. Heming, "K-means clustering algorithms: A comprehensive review, variants analysis, and advances in the era of big data," *Inf. Sci.*, vol. 622, pp. 178–210, Apr. 2023.

[36] S. E. Hashemi, F. Gholian-Jouybari, and M. Hajiaghaei-Keshteli, "A fuzzy C-means algorithm for optimizing data clustering," *Expert Syst. Appl.*, vol. 227, Oct. 2023, Art. no. 120377.

[37] M. Shutaywi and N. N. Kachouie, "Silhouette analysis for performance evaluation in machine learning with applications to clustering," *Entropy*, vol. 23, no. 6, p. 759, Jun. 2021.

[38] F. Ros, R. Riad, and S. Guillaume, "PDBI: A partitioning davies-bouldin index for clustering evaluation," *Neurocomputing*, vol. 528, pp. 178–199, Apr. 2023.

[39] Y. Lv, S. Liu, Y. Gao, J. Dai, Z. Ren, and Y. Liu, "An ultra-wideband indoor localization algorithm with improved cubature Kalman filtering based on sigmoid function," *Appl. Sci.*, vol. 14, no. 6, p. 2239, Mar. 2024.

[40] B. Morawska, P. Lipinski, K. Lichy, and K. Adamkiewicz, "Transfer learning-based UWB indoor localization using MHT-MDC and clusterization-based sparse fingerprinting," *J. Comput. Sci.*, vol. 61, May 2022, Art. no. 101654.

[41] K. Paszek and D. Grzechca, "Using the LSTM neural network and the UWB positioning system to predict the position of low and high speed moving objects," *Sensors*, vol. 23, no. 19, p. 8270, Oct. 2023.

[42] Q. Liu, Z. Yin, Y. Zhao, Z. Wu, and M. Wu, "UWB LOS/NLOS identification in multiple indoor environments using deep learning methods," *Phys. Commun.*, vol. 52, Jun. 2022, Art. no. 101695.

[43] J. Sun, W. Shi, Z. Yang, J. Yang, and G. Gui, "Behavioral modeling and linearization of wideband RF power amplifiers using BiLSTM networks for 5G wireless systems," *IEEE Trans. Veh. Technol.*, vol. 68, no. 11, pp. 10348–10356, Nov. 2019.

[44] B. Jia, J. Liu, T. Feng, B. Huang, T. Baker, and H. Tawfik, "TTSL: An indoor localization method based on temporal convolutional network using time-series RSSI," *Comput. Commun.*, vol. 193, pp. 293–301, Sep. 2022.

[45] C. Jiang, S. Chen, Y. Chen, D. Liu, and Y. Bo, "An UWB channel impulse response de-noising method for NLOS/LOS classification boosting," *IEEE Commun. Lett.*, vol. 24, no. 11, pp. 2513–2517, Nov. 2020.



Shixun Wu was born in Hubei, China. He received the B.S. degree from Hubei University of Arts and Science, Xiangyang, China, the M.S. degree from Central China Normal University, Wuhan, China, and the Ph.D. degree from the Department of Electrical and Computer Engineering, Central China Normal University, Wuhan, in 2006, 2009, and 2012, respectively.

He is currently a Vice-Professor with the College of Information Science and Engineering, Chongqing Jiaotong University, Chongqing, China. He is also the Vice-Dean of the Department of Communication Engineering. His current research interests include wireless localization, neural network localization, and wireless communication.



Xiao Wang was born in 1999. He received the B.Sc. degree in communication engineering from Chongqing Jiaotong University, Chongqing, China, in 2021, where he is currently pursuing the M.Sc. degree in computer science and technology.

His research primarily focuses on ultraw-ideband (UWB) indoor positioning and deep learning applications.

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:29:01 UTC from IEEE Xplore. Restrictions apply.



Miao Zhang (Member, IEEE) received the B.Sc. degree in optical information science and technology from Guizhou University, Guiyang, China, the M.Sc. degree in communications and signal processing from the University of Newcastle upon Tyne, Newcastle upon Tyne, U.K., and the Ph.D. degree from the University of York, York, U.K., in 2011, 2015 and 2020, respectively.

He is currently an Associate Professor at the School of Information Science and Engineering, Chongqing Jiaotong University, Chongqing, China. His research interests are convex optimization techniques, intelligent reflecting surface-assisted wireless networks, physical layer security, and machine learning techniques for wireless communications.



Zheng Chu (Member, IEEE) received the M.Sc. and Ph.D. degrees from Newcastle University, Newcastle upon Tyne, U.K., in 2012 and 2016, respectively.

He held positions with the University of Surrey, Guildford, U.K., from 2017 to 2023, and Middlesex University, from 2016 to 2017, respectively. He is an Assistant Professor with the Department of Electrical and Electronic Engineering (EEE), University of Nottingham Ningbo China (UNNC), Ningbo, China. His current research interests include smart radio environments/smart reflecting surfaces, 6G communication networks, IoT networks, AI-driven future networks, wireless security, and wireless-powered networks.

Dr. Chu has been included in the list of world top 2% scientists for the term of 2020–2023, received the Exemplary Reviewer for IEEE TRANSACTIONS ON COMMUNICATIONS in 2022, and the Best Paper Award of International Conference on Ubiquitous Communications Communication in 2024.



Zhangli Lan was born in 1973. He received the B.S. degree in communication from the Electronic Engineering College, Hefei, China, in 1997, and the M.S. degree in computer system structure and the Ph.D. degree in computer software and theory from Chongqing University, Chongqing, China, in 2004 and 2008, respectively.

Since 2011, he has been a Professor with the School of Information Science and Engineering, Chongqing Jiaotong University, Chongqing. His current research interests include digital image processing, pattern recognition, and traffic information processing.



Kai Xu was born in 1970. He received the M.S. degree in control theory and control engineering from Chongqing University, Chongqing, China, in 2006.

From 1993 to 2003, he was a Senior Engineer with Xichang Satellite Launching Center, Xichang, China. Since 2008, he has been a Professor with the College of Information Science and Engineering, Chongqing Jiaotong University. Prof. Xu is an Expert Commissioner of Chongqing Science and Technology Commission and Chongqing Construction Commission. His current research interests include fuzzy control, adaptive control, and intelligent algorithms.



Shuang Jin received the B.S. degree in automation from Chongqing University of Posts and Telecommunications, Chongqing, China, in 2015, and the Ph.D. degree in control theory and control engineering from Chongqing University, Chongqing, in 2021.

He is currently a Lecturer with the College of Information Science and Engineering, Chongqing Jiaotong University, Chongqing. His research focuses on intelligent transportation systems, connected and automated vehicles, and cooperative control.

Authorized licensed use limited to: Viet Nam National University Ho Chi Minh City. Downloaded on January 05,2026 at 01:29:01 UTC from IEEE Xplore. Restrictions apply.